

Mesoscale Accessibility, Basin Transport, and Memory in Adaptive Landscape Systems

Abstract

We introduce a mesoscale accessibility observable for adaptive landscape systems and examine how landscape reorganization emerges through basin-to-basin transport under coarse-graining. The model combines mutation-driven exploration, selection-mediated localization, and local density competition. Population snapshots are mapped to mesoscale basin states, and accessibility is measured using a bounded transport-based observable derived from empirical basin-to-basin flux.

Across mutation scans, the model exhibits a restricted mutation band in which inter-basin transport activates. This accessibility activation separates from realized diversity: diversity measures occupation, whereas accessibility measures the possibility of sustained movement between basin states. The onset of accessibility leaves path-dependent basin memory under upward and downward mutation sweeps, and this memory signature persists under an effective density-field coarse graining.

The resulting transition band remains concentrated across a finite family of coarse-graining operators and persists under finite population-size variation, while post-threshold filling capacity remains population-size dependent. Null and shuffle controls show that fitted threshold locations can persist even when temporal, ordering, or structural signatures fail. These results support a transport-based statistical-mechanical view of adaptive landscape reorganization: accessibility activation is a mesoscale dynamical process involving basin transport, memory, and robustness under coarse-graining, rather than a diversity threshold alone.

1. Introduction

Adaptive biological systems reorganize under mutation, selection, ecological interaction, and internal constraints. Such reorganization may involve changes in occupied states, changes in the relations among states, or changes in the paths by which a system can continue to explore its state space. Understanding how organization changes in adaptive systems has motivated work on adaptive landscapes, biological organization, complex adaptive systems, and systems-

theoretic descriptions of evolution [1,2,22-27]. The present paper addresses this conversation at a deliberately minimal level: how can one measure the onset of adaptive reorganization in an adaptive landscape model?

Many descriptions of adaptive reorganization are occupancy based. Diversity, for example, measures realized occupation: how many niches, basins, or states are populated at a given time. Adaptive-dynamics formulations describe how mutation, selection, and ecological feedback can produce branching or diversification [22-24], while broader systems approaches emphasize adaptive organization, model structure, and system-level behavior [25-30]. These perspectives leave a measurement gap. Occupancy does not directly indicate whether persistent movement between occupied or nearby states has become possible. Two systems may have similar diversity but different state-space connectivity, and a system may begin to open movement pathways before it fully fills the accessible state space.

This motivates the introduction of a transport-based mesoscale observable for adaptive organization. We introduce this object through basin accessibility. In a minimal adaptive landscape model, agents reproduce under selection, experience local density competition, and explore phenotype space through mutation-driven diffusion. Population snapshots are mapped to mesoscale basin states. From the resulting occupancy trajectories, we construct basin-to-basin flux and define an accessibility order parameter $\Psi = J / (1 + J)$, where J is basin-to-basin flux. In this framing, diversity measures occupancy, whereas accessibility measures sustained transport between mesoscale basin states. The construction is related in spirit to coarse-grained descriptions of slow or metastable dynamics, but is used here as an operational measurement framework rather than as an exact reduced dynamics [11-17].

The setting considered here combines mutation-driven exploration, selection-mediated localization, local density competition, and history-dependent occupation. In such systems, increasing mutation strength can change both the amount of occupied state space and the transport structure between coarse ecological states. A fitted threshold in a diversity curve therefore provides only partial information: it may locate a response boundary, but it does not by itself reveal whether a coherent basin-transport process has emerged.

The central question of the paper is: how does basin-to-basin accessibility emerge in adaptive landscape dynamics, and does this emergence leave a coherent mesoscopic signature under coarse-graining? To answer this, we combine agent-based simulations, an effective density-field description, basin-to-basin flux

analysis, hysteresis tests, operator-stability checks, finite-size variation, and null and shuffle controls.

We find that mutation controls a restricted transition band separating localized occupation from multi-niche occupation. Accessibility flux activates within this band, separates from realized diversity, and leaves basin memory under mutation sweeps. The transition band remains concentrated across tested coarse-graining operators and finite-size conditions, while post-threshold filling capacity depends on population size. Finally, null and shuffled controls show that a fitted threshold alone is insufficient: apparent threshold locations can persist while structural, temporal, or ordering diagnostics fail.

The contribution is deliberately bounded. We do not claim a universal critical exponent, a strict RG fixed point, or a general theory of all adaptive systems. Instead, we define a reproducible framework for measuring accessibility activation, basin memory, and robustness under coarse-graining in a minimal adaptive landscape system. Diagnostic controls are used to verify that the observed transport signature is not merely a fitted-threshold artifact.

2. Results

2.1 Restricted mutation band for diversification

We first asked whether the minimal adaptive landscape model exhibits a reproducible diversification onset. The model combines mutation-driven exploration, selection toward niche structure, reproduction proportional to fitness, and local density-dependent competition (Fig. 1A,B). The measured macroscopic observable is the diversity order parameter ϕ , defined from the effective number of occupied niches.

Scanning mutation strength V and selection strength S revealed a restricted transition band separating low-occupancy and high-occupancy regimes (Fig. 1C). At low mutation, populations remain localized within a small number of basins. As mutation increases through a narrow interval, the system shifts toward multi-niche occupation. Beyond this interval, diversity approaches a high occupancy regime across a broad range of S .

The diversity response boundary $V_{\phi}^*(S)$ was extracted from the strongest response of ϕ with respect to mutation strength. Across the explored selection range, $V_{\phi}^*(S)$ remained confined to a narrow mutation interval (Fig. 1D). This indicates that diversification onset is not distributed broadly across mutation values but is associated with a restricted mutation band. Figure 1 therefore

defines the first empirical object of the study: a mutation-controlled diversity response boundary in the adaptive landscape system.

2.2 Accessibility activates within the transition band

The existence of a diversity response does not by itself identify what dynamical object is changing. Diversity measures realized occupancy: how many mesoscale basin states are populated. The transition, however, may depend more directly on whether the system can move between such states. We therefore constructed mesoscale basin states and measured accessibility as sustained inter-basin transport, with J as the corresponding basin-to-basin flux (Fig. 2A,B).

Below the transition band, coarse-grained trajectories are dominated by self-transition within a low-occupancy basin state. Near the accessibility threshold V_ψ^* , persistent inter-basin transport begins to appear. Above the transition band, multiple transport corridors become active (Fig. 2B). This schematic description is supported by the basin-to-basin flux J , which increases sharply within the same mutation interval identified from the diversity response boundary (Fig. 2C).

The accessibility order parameter Ψ separates from the diversity observable ϕ near the transition (Fig. 2D). ϕ measures realized occupancy, whereas Ψ measures the possibility of sustained inter-basin transport. The two quantities therefore describe different levels of organization: diversity is a state observable, while accessibility is a dynamical transport observable. This distinction motivates the central interpretation of the transition as an accessibility activation rather than merely an increase in occupied niche count.

2.3 Basin memory and mesoscopic survival

If accessibility activation is a dynamical transition, it should leave evidence beyond a static response curve. We therefore tested whether the system exhibits path dependence under upward and downward mutation sweeps. In the microscopic agent-based model, the up-sweep and down-sweep trajectories separate near the accessibility band, indicating hysteresis and basin memory (Fig. 3A).

We next asked whether this memory survives coarse-graining. A deterministic mesoscopic PDE description preserves the qualitative hysteresis structure (Fig. 3B). The amplitudes of memory need not match exactly between the finite-agent ABM and the deterministic density-field description, because finite-population stochasticity and extinction can soften or reshape memory. For this reason, we compared normalized memory signatures rather than claiming quantitative

equality of hysteresis magnitude. Both descriptions concentrate memory near the accessibility band (Fig. 3C).

Finally, we compared accessibility onset with a local stability diagnostic. The leading stability eigenvalue does not predict the observed accessibility onset by itself (Fig. 3D). This does not imply that local stability is irrelevant; rather, it indicates that local stability alone is insufficient to account for the observed transition. Together, the hysteresis, coarse-grained survival, and stability mismatch support the interpretation that accessibility activation is a dynamical basin-memory phenomenon.

2.4 Robustness without scale invariance

Complex-systems transitions can be sensitive to the representation used to define states and measure changes between them. To test whether the accessibility band is an artifact of a single coarse-graining choice, we applied multiple operator families to the same underlying occupancy trajectories (Fig. 4A). These operators represent alternative geometric descriptions rather than sequential steps in a single pipeline.

Across operator families, the estimated transition locations concentrate within a common mutation band (Fig. 4B). The operators do not define identical observational mesoscales, but they recover a shared transition interval. This supports operator stability of the transition band while avoiding the stronger claim of strict scale invariance.

We also tested finite-size persistence by varying population size N . Estimated transition locations remained confined to a stable interval with overlapping bootstrap confidence intervals (Fig. 4C). In contrast, maximum filling capacity increased with N (Fig. 4D). This separation supports a two-level interpretation: accessibility onset is comparatively stable, whereas post-onset filling depends on finite population capacity. Thus, robustness appears at the transition-band level, not as a claim of scale-free critical invariance.

2.5 Diagnostic consistency controls

The preceding results show that a candidate accessibility threshold is reproducible, dynamical, and robust across several perturbations. However, a fitted threshold alone is not sufficient evidence for a coherent transition candidate. Nonlinear curves, noisy observables, and shuffled trajectories can all produce apparent threshold locations. We therefore evaluated candidate

accessibility activation using a diagnostic consistency protocol designed to test against multiple classes of threshold artifact (Fig. 5A).

Physical null models reduce or remove key mechanisms and thereby test whether the observed transition structure depends on the adaptive landscape dynamics (Fig. 5B). Shuffle controls provide a different challenge: they can preserve some fitted threshold-like behavior while disrupting temporal or mutation-order structure (Fig. 5C). These controls show that the existence of an estimated V^* is necessary for analysis but not sufficient for interpretation as a coherent accessibility transition.

The full diagnostic matrix evaluates six conditions simultaneously: accessibility activation, peak localization, operator agreement, coherent filling, temporal integrity, and mutation-order consistency (Fig. 5D). Only the baseline system passes all six diagnostics. Null and shuffled controls fail in distinct ways, indicating that the protocol is not merely selecting for a single metric but instead tests several independent failure modes. Figure 5 therefore serves as a consistency control for the basin-transport interpretation: a candidate transition is interpreted as diagnostically consistent only when it survives artifact-oriented diagnostics.

2.6 Diagnostic rationale

The six diagnostics are not proposed as a necessary-and-sufficient theorem for all adaptive systems. They form a minimal operational protocol for this model class, chosen to reject distinct artifacts that can mimic threshold-like behavior.

Accessibility activation tests against transient fluctuations without sustained growth in inter-basin transport. **Peak localization** tests against thresholds chosen from gradually varying or noisy curves. **Operator agreement** tests against coarse-graining-specific transitions. **Coherent filling** tests against isolated jumps in derived statistics without structured occupation. **Temporal integrity** tests against thresholds that persist after trajectory order is destroyed. **Mutation-order consistency** tests against boundaries induced by rearranging or pooling mutation conditions.

Together, these diagnostics define a conservative artifact-rejection framework for coherent accessibility activation. The goal is to reduce false positives in threshold detection, not to claim a universal diagnostic set.

3. Methods

3.1 Agent-based adaptive landscape model

Agents occupy a continuous phenotype space, with agent i represented by a state vector $x_i \in \mathbb{R}^d$. Unless otherwise stated, simulations use a two-dimensional phenotype space. The system contains E niche centers $\mu_1, \dots, \mu_E$, which define the ecological landscape.

At each generation, agents are assigned ecological fitness from three ingredients: attraction to niche structure, local density competition, and mutation-driven exploration after reproduction. The base niche fitness is computed from the distance to the closest niche:

$$U(x_i) = \min_k \|x_i - \mu_k\|^2,$$

Equation (1)

$$F_{base}(x_i) = \exp[-S U(x_i)],$$

Equation (2)

where S is the selection strength. Local density is estimated from the neighborhood of each agent, using a finite interaction scale such as k -nearest neighbors. Competition enters as a density-dependent penalty:

$$F_i = \frac{F_{base}(x_i)}{1 + \lambda \rho(x_i)},$$

Equation (3)

where λ controls local competition strength and $\rho(x_i)$ is the estimated local density around agent i .

Reproduction is fitness-proportional. A new generation is sampled from the current population with probabilities proportional to F_i . Mutation is then applied as Gaussian displacement:

$$x'_i = x_{parent} + \epsilon, \epsilon \sim N(0, V I).$$

Equation (4)

Here V denotes mutation strength as a diffusion variance scale, not a mutation probability. This distinction is important because V controls exploration distance in phenotype space rather than the fraction of agents that mutate.

3.2 Mesoscopic coarse-graining

Each population snapshot is mapped to a niche-occupancy vector:

$$p_t \in \Delta^{E-1}.$$

Equation (5)

where $p_t[k]$ is the fraction of agents assigned to niche k , typically by nearest-neighbor assignment. The sequence $\{p_t\}$ forms the occupancy trajectory used for downstream coarse-graining.

In this manuscript, a basin is an operational mesoscale state induced by a coarse-graining operator. Basins are analytical measurement objects rather than predefined ecological entities or exact microscopic attractors. A coarse-graining operator maps occupancy vectors to either discrete basin states or soft memberships:

$$C_\epsilon: \Delta^{E-1} \rightarrow M_\epsilon.$$

Equation (6)

For hard coarse-graining, the output is a discrete state label:

$$s_t = C_\epsilon(p_t).$$

Equation (7)

For soft coarse-graining, the output is a membership vector over prototype or landmark states:

$$w_t = C_\epsilon(p_t), \sum_k w_t[k] = 1.$$

Equation (8)

This representation provides the mesoscopic state space on which accessibility and basin flux are measured. It follows the general logic of coarse-graining high-dimensional dynamics into slower collective or metastable descriptions, while remaining deliberately operational for the present adaptive landscape model [11-15].

3.3 Diversity and accessibility observables

The diversity observable ϕ is based on effective niche occupation. Given a niche-occupancy vector p , the effective number of occupied niches is computed from Shannon entropy:

$$H(p) = - \sum_k p_k \log p_k, N_{eff} = \exp[H(p)], \phi = \frac{N_{eff}}{E}.$$

Equation (9)

Thus ϕ measures realized occupancy: how broadly the population fills the available niche set.

Accessibility is measured from transitions between mesoscale basin states. For hard states, transition counts are:

$$N_{ij} = \# \{t : s_t = i, s_{t+1} = j\}.$$

Equation (10)

For soft states, transition mass is accumulated as:

$$N_{ij} = \sum_t w_t[i] w_{t+1}[j].$$

Equation (11)

The row-normalized transition matrix is:

$$P_{ij} = \frac{N_{ij}}{\sum_j N_{ij}}.$$

Equation (12)

and the empirical weight of basin i is:

$$\pi_i = \frac{\sum_j N_{ij}}{\sum_{i,j} N_{ij}}.$$

Equation (13)

Inter-basin flux is defined as the probability mass leaving its current basin:

$$J = \sum_i \pi_i \sum_{j \neq i} P_{ij}.$$

Equation (14)

equivalently:

$$J = 1 - \sum_i \pi_i P_{ii}.$$

Equation (15)

The bounded accessibility order parameter is:

$$\Psi = \frac{J}{1 + J}.$$

Equation (16)

The transformation in Eq. (16) does not introduce additional structure; it is used only to obtain a bounded observable on $[0, 1)$ while preserving monotonicity with respect to J .

In the terminology of the manuscript, ϕ measures occupancy, whereas Ψ measures the possibility of sustained inter-basin transport.

3.4 Threshold and response-boundary estimation

Mutation-response boundaries were estimated from response curves over mutation strength V . For diversity scans, the diversity response boundary is obtained from the largest numerical derivative of ϕ with respect to V :

$$V_{\phi}^* = \arg \max_V \frac{d\phi}{dV}.$$

Equation (17)

For accessibility analyses, the primary threshold is defined from the accessibility order parameter:

$$V_{\Psi}^* = \arg \max_V \frac{d\Psi}{dV}.$$

Equation (18)

Unless otherwise specified, V_{Ψ}^* is the primary accessibility threshold used for operator-stability and diagnostic-consistency analyses. Diversity-response thresholds and fluctuation-based checks are treated as consistency measures rather than independent definitions of the transition.

3.5 Alternative coarse-graining operators

To test whether the accessibility band depends on a particular representation, the same occupancy trajectories were analyzed using a finite family of coarse-graining operators. The tested families include:

- hard occupancy threshold
- Euclidean kernel
- cosine kernel
- Hellinger kernel
- Wasserstein / transport kernel
- spectral embedding
- diffusion-map kernel

These operators provide alternative mesoscopic geometries of the same underlying trajectories. The analysis does not claim invariance under all possible operators; robustness claims are restricted to the tested operator family. Each operator can select its own observational scale, but the key quantity compared across operators is the inferred transition band in mutation space. This operator-comparison logic is motivated by the broader use of spectral, kernel, and Markov-state coarse-graining methods for extracting low-dimensional structure from complex trajectories [13-17].

3.6 Bootstrap uncertainty estimation

Uncertainty in transition estimates was quantified by bootstrapping over independent simulation repeats. The bootstrap unit is the independent replicate, not a time window within a single trajectory, because trajectory windows are temporally correlated.

For each operator family, the protocol is:

1. Run K independent simulation repeats for each mutation value.
2. Compute V^* for each independent repeat or replicate-level summary.
3. Resample the replicate set with replacement.
4. Recompute the operator-level mean V^* for each bootstrap sample.
5. Report the bootstrap mean and 95% confidence interval.

Where shown, error bars indicate bootstrap 95% confidence intervals. Operator sensitivity is summarized by the coefficient of variation of operator-mean thresholds:

$$R_v = \frac{\text{std}_a(\text{mean}_k V_{a,k}^*)}{\text{mean}_a(\text{mean}_k V_{a,k}^*)}.$$

Equation (19)

where a indexes coarse-graining operators and k indexes independent simulation repeats. Smaller R_v indicates stronger concentration of operator-selected thresholds.

3.7 Null and shuffle controls

Controls were divided into physical null models and shuffle controls because they test different failure modes.

Physical nulls remove or weaken model mechanisms. The main physical controls include:

- $\lambda=0$.
- no-selection / flat-like landscape control.

The $\lambda=0$ control removes local density competition. The no-selection or flat-like landscape control removes structured localization to ecological basins. These controls test whether the observed transition structure depends on the adaptive landscape mechanisms.

Shuffle controls preserve some marginal features of the data while disrupting ordering structure. They ask whether the inferred transition depends on a coherent path through state space and on the ordered relation between mutation strength and system response. The main shuffle controls are:

- time-shuffled occupancy trajectories
- mutation-label shuffled trajectories

Time shuffling preserves occupancy frequencies while destroying dynamical path continuity. It therefore tests whether accessibility activation depends on coherent temporal evolution rather than on static state distributions. Mutation-label shuffling preserves measured observables while disrupting the ordered relationship between mutation strength and observed occupancy or accessibility structure. It therefore tests whether the inferred transition reflects parameter-driven reorganization rather than an accidental threshold fit. These controls test whether fitted thresholds can survive when temporal or parameter-ordering structure has been removed.

3.8 Diagnostic consistency protocol

A fitted V^* is treated as necessary for threshold analysis but not sufficient for interpreting an observed response as a coherent accessibility-transition candidate. We therefore use six diagnostics as a conservative consistency framework:

1. accessibility activation
2. peak localization
3. operator agreement
4. coherent filling
5. temporal integrity
6. mutation-order consistency

The diagnostics are grouped by the failure mode they test:

Diagnostic class	Diagnostic	Failure mode tested
primary response	accessibility activation	fluctuation artifact
primary response	peak localization	threshold-placement artifact
robustness	operator agreement	representation artifact
robustness	coherent filling	isolated-jump artifact
control	temporal integrity	time-order artifact
control	mutation-order consistency	ordering artifact

Operational diagnostic quantities include:

Metric	Operational definition
S_{range}	dynamic range of Ψ over the mutation scan
S_{peak}	response prominence, $\max(d\Psi/dV)/std(d\Psi/dV)$
S_{op}	operator-consistency proxy, $1/R_V$
S_{fill}	filling proxy, maximum observed

Metric	Operational definition
	diversity ϕ_{max}
$S_{temporal}$	inverse mean occupancy step distance, $1 / mean \ p_{t+1} - p_t\ $
S_{order}	correlation between mutation strength and diversity response, $corr(V, \phi)$

Consistency thresholds are applied in the same way to the baseline model and all controls. The protocol is intentionally conservative: failure of any diagnostic indicates that the evidence is insufficient to interpret the fitted threshold as a coherent accessibility-transition candidate. Results 2.5 and 2.6 describe the rationale for these diagnostics; this section defines their operational use.

4. Discussion

4.1 Basin Transport and Accessibility

This study examines adaptive organization through a minimal adaptive landscape model in which occupation and transport can be distinguished as complementary observables. The transition appears first as a restricted mutation band separating localized occupation from multi-niche occupation, but the central object is not diversity alone. By coarse-graining occupancy trajectories into basin states, we find that mutation controls the onset of sustained inter-basin transport. This accessibility activation is accompanied by basin memory, persists qualitatively in a mesoscopic density-field description, remains concentrated across a finite family of coarse-graining operators, and survives finite-size variation as a transition band.

The main contribution is therefore a transport-based description of adaptive landscape reorganization. The model provides a measurable accessibility order parameter that distinguishes dynamical transport between basin states from realized niche occupation. Diagnostic controls then test whether the observed transport signature survives perturbations that preserve some threshold-like features while destroying temporal, ordering, or structural coherence. In this sense, the paper shifts the object of inference from diversity change alone to basin-to-basin accessibility under coarse-graining.

This framing is intentionally narrower than a general theory of adaptive systems or biological organization. The results show that, in this model class, a transition can be identified at the level of accessibility structure even when exact

observational scale and post-threshold filling vary. The stable object is the transition band and its diagnostic signature, not a single universal scale or exponent.

4.2 Accessibility and Memory

A central implication for adaptive organization is that accessibility and diversity should not be treated as interchangeable observables. Diversity measures realized occupation, whereas accessibility measures whether transport between mesoscale basin states can be sustained. This distinction matters because an organized adaptive state is not described only by what is occupied, but also by which movements between coarse states remain dynamically accessible. Accessibility can activate before realized diversity has fully changed, while high occupancy alone does not guarantee coherent dynamical access between basin states.

In the present model, mutation opens basin-to-basin transport pathways before ecological filling is complete. This separates onset from filling: the accessibility order parameter captures the onset of transport, whereas the diversity order parameter captures downstream niche occupation. The finite-size results reinforce this interpretation, since transition onset remains comparatively stable while maximum filling capacity grows with population size. A diversity-only description can reveal that something changes, but not whether the change reflects coherent movement, random diffusion, or metric-dependent clustering.

The hysteresis results further show that accessibility activation is not only a static change in a response curve. Upward and downward mutation sweeps leave different basin-occupation histories, and the normalized memory signature persists in an effective density-field description. The comparison does not claim equality of microscopic and mesoscopic hysteresis magnitudes. Rather, it shows that the accessibility transition leaves a coarse-grained memory trace, consistent with a dynamical reorganization of basin transport.

4.3 Robustness of Mesoscopic Dynamics

The robustness results should be read as mesoscopic consistency rather than as a claim of scale-free universality. Different coarse-graining operators impose different geometries on the same occupancy trajectories, yet the inferred transition locations concentrate within a shared mutation band. Similarly, finite population-size variation separates onset from filling: accessibility onset remains comparatively stable, whereas post-onset occupation capacity increases with population size.

The diagnostic controls provide an additional check on this interpretation. A fitted threshold is necessary for threshold analysis but insufficient as evidence for a transport-mediated transition. This concern is consistent with broader work on critical transitions and early-warning signals, where apparent thresholds may depend on observation, noise, model structure, or the diagnostic used [4,18-20]. In the present setting, coarse-graining choices, clustering rules, and metric definitions add an additional representation layer. Null and shuffle controls show directly that fitted threshold-like locations can persist even when temporal structure or mutation-order coherence has been disrupted. The diagnostic protocol is therefore used as an artifact-rejection control for the observed accessibility activation, not as a general theorem of transition identification.

The observed accessibility onset is also not associated with a detected local bifurcation in the present analysis. This supports treating the onset as a basin-transport reorganization rather than as a transition identified from local stability alone.

4.4 Limits

The claims of this manuscript are deliberately bounded. The model is minimal and stylized. It is designed to expose relationships among mutation-driven exploration, selection-mediated localization, local competition, and coarse-grained accessibility. It is not intended as a full ecological model, a complete evolutionary theory, or a direct explanation of any particular biological system.

The robustness claim is also limited. The transition band is stable across the tested operator families and finite-size conditions, but this does not imply invariance under all possible coarse-graining operators. Different operators can select different observational mesoscales. The claim is that the tested operators recover a shared mutation band, not that they converge to a single scale-free description. The results support transition-band robustness without requiring scale-free invariance.

The diagnostic consistency protocol is operational rather than necessary and sufficient. The six diagnostics were chosen to address the artifact classes exposed by this model and its controls. Other adaptive systems may require additional diagnostics or modified criteria. Similarly, the exponent structure is not claimed to be universal, and no strict RG fixed point is claimed. The analysis is best interpreted as a reproducible mesoscale framework for measuring accessibility activation in a defined class of adaptive landscape systems.

Finally, the mesoscopic PDE and stability analysis should be read as effective coarse-grained probes, not as exact derivations of the microscopic process. The local stability mismatch suggests that accessibility onset is not captured by local stability alone, but it does not imply that stability analysis is irrelevant. Rather, it indicates that transport through basin structure and path-dependent memory are essential parts of the observed transition signature. This is why the manuscript treats hysteresis and basin memory as transition diagnostics rather than relying only on local stability or derivative-based threshold estimates [3,4].

4.5 Outlook

Several extensions follow naturally from this framework. First, stochastic PDE descriptions could clarify how finite-population noise modifies memory, extinction, and hysteresis relative to deterministic density-field dynamics. Second, barrier or escape-rate estimates could connect accessibility flux to activated transport across basin boundaries, providing a more analytical description of the transition band [21]. Third, future work may examine the extent to which the observed accessibility activation persists under alternative landscape geometries, mutation kernels, and competition structures.

These directions should be treated as future validation work rather than as claims of generality. The present contribution is narrower: it defines a measurable accessibility object, shows how basin-to-basin transport activates and leaves memory in a minimal adaptive landscape system, and uses diagnostic controls to separate coherent accessibility activation from threshold-like artifacts. The central object of the present study is therefore not diversity itself but accessibility as a transport-based mesoscale observable. From this perspective, adaptive organization can be studied by separating occupancy-based statistics from transport-based observables that capture state-space connectivity and basin movement under explicit coarse-graining.

Figure Captions

Figure 1 | Restricted mutation band for diversification

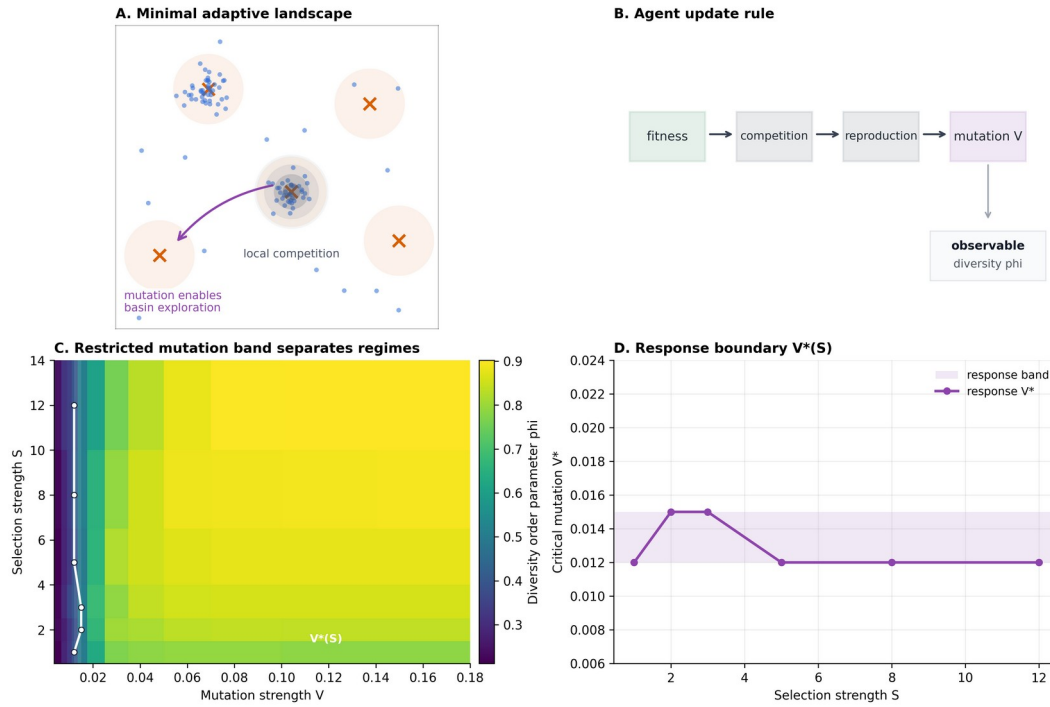


Figure 1 | Restricted mutation band for diversification

A restricted mutation band separates localized occupation from multi-niche diversification. (A) Minimal adaptive landscape used in the agent-based model. Mutation enables exploration between basins, whereas local competition is shown as a density-dependent pressure around occupied regions. (B) Agent update rule: fitness, competition, reproduction, and mutation jointly determine the next population state and the measured diversity observable ϕ . (C) Diversity order parameter ϕ across mutation strength V and selection strength S . A narrow mutation interval separates low-occupancy and high-occupancy regimes over a broad range of selection strengths. (D) Diversity response boundary $V_\phi^*(S)$, showing that diversification onset remains confined to a restricted mutation band across the explored parameter range. Together, these results support the existence of mutation-controlled diversification onset in the minimal adaptive landscape system.

Figure 2 | Accessibility as the hidden transition variable

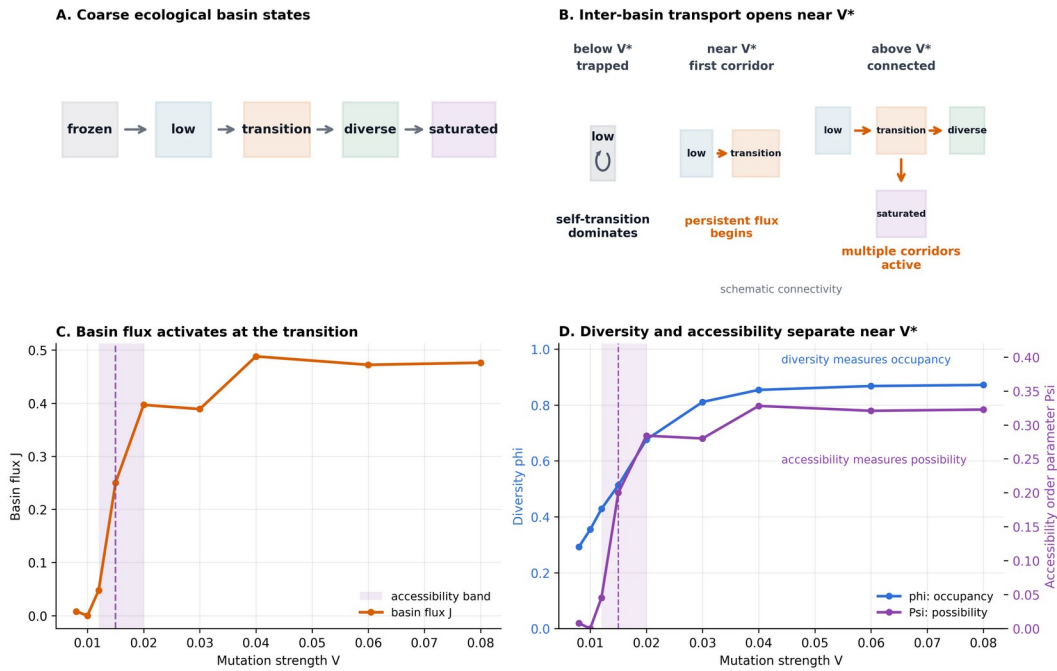


Figure 2 | Accessibility as the hidden transition variable

Accessibility activates within the restricted mutation band. (A) Mesoscale basin states used to describe mesoscopic occupancy structure. States represent coarse classes of niche occupation rather than individual microscopic configurations. (B) Schematic connectivity between basin states below, near, and above the transition region. Below the transition band, self-transition dominates; near the accessibility threshold V_ψ^* , persistent inter-basin transport emerges; above the band, multiple transport corridors become active. Links in this panel are schematic and indicate accessibility opening rather than exact transition-rate estimates. (C) Basin-to-basin flux J as a function of mutation strength V . Accessibility activation occurs within the mutation interval identified by the diversity response boundary. (D) Diversity ϕ and accessibility order parameter Ψ separate near the transition: ϕ measures realized occupancy, whereas Ψ measures the possibility of sustained inter-basin transport. These results identify accessibility activation as the mesoscopic process underlying diversification onset.

Figure 3 | Basin memory and mesoscopic survival

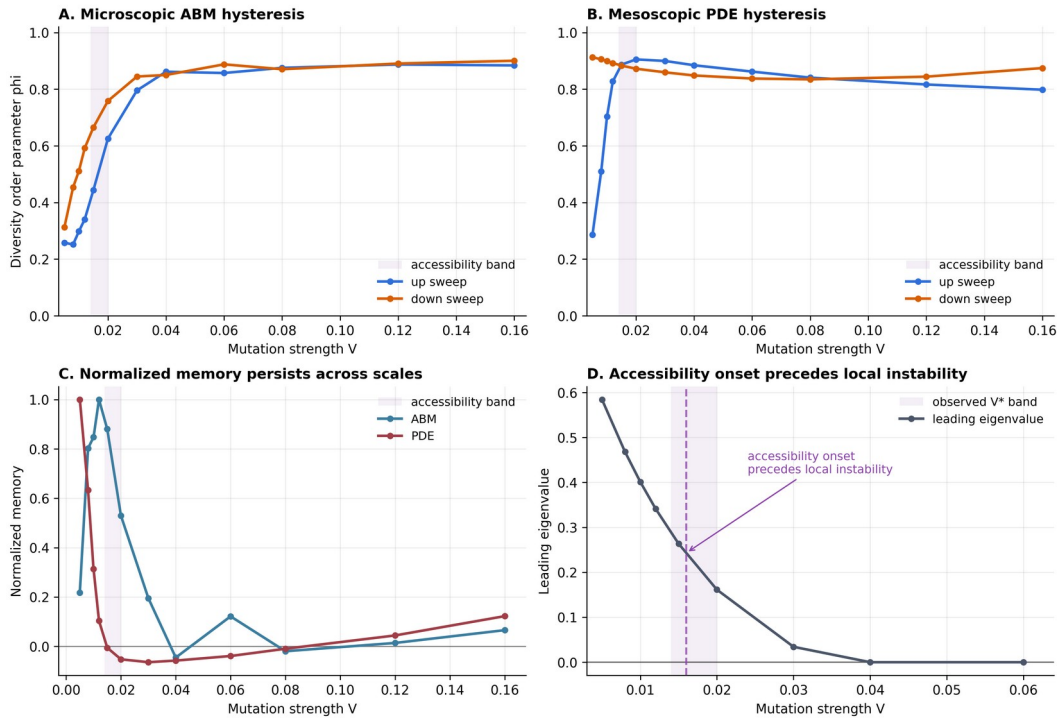


Figure 3 | Basin memory and mesoscopic survival

Accessibility activation leaves persistent basin memory. (A) Hysteresis in the microscopic agent-based model under upward and downward mutation sweeps. Distinct trajectories persist near the transition region. (B) Corresponding hysteresis in the coarse-grained PDE description. (C) Normalized memory signatures from the ABM and PDE descriptions. Although absolute amplitudes may differ, both descriptions concentrate memory near the accessibility band. (D) Leading stability eigenvalue as a function of mutation strength. The observed accessibility onset is not predicted by local stability alone. The persistence of hysteresis and memory signatures across descriptions indicates that accessibility activation is a dynamical transition with measurable basin memory.

Figure 4 | Robustness without scale invariance

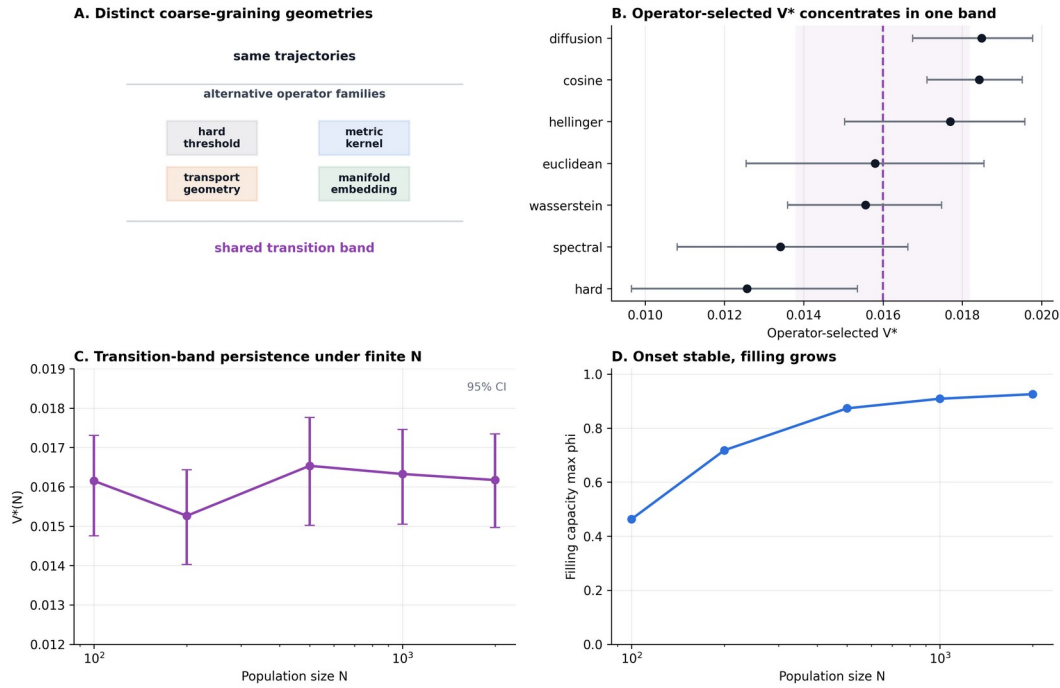


Figure 4 | Robustness without scale invariance

The accessibility transition band remains stable across coarse-graining operators and finite population sizes. (A) Distinct coarse-graining operator families applied to the same underlying occupancy trajectories. Operators represent alternative geometric descriptions, not sequential analysis steps. (B) Operator-selected transition locations V^* recovered from multiple coarse-graining schemes. Despite operator-dependent representations, transition estimates concentrate within a common mutation band. (C) Estimated transition location as a function of population size N . Error bars denote bootstrap 95% confidence intervals. Transition onset remains confined to a stable interval across finite-size systems. (D) Maximum filling capacity as a function of population size. Larger systems support greater occupancy while transition onset remains comparatively stable. These results support transition-band robustness without requiring scale-free invariance.

Figure 5 | Diagnostic consistency analysis

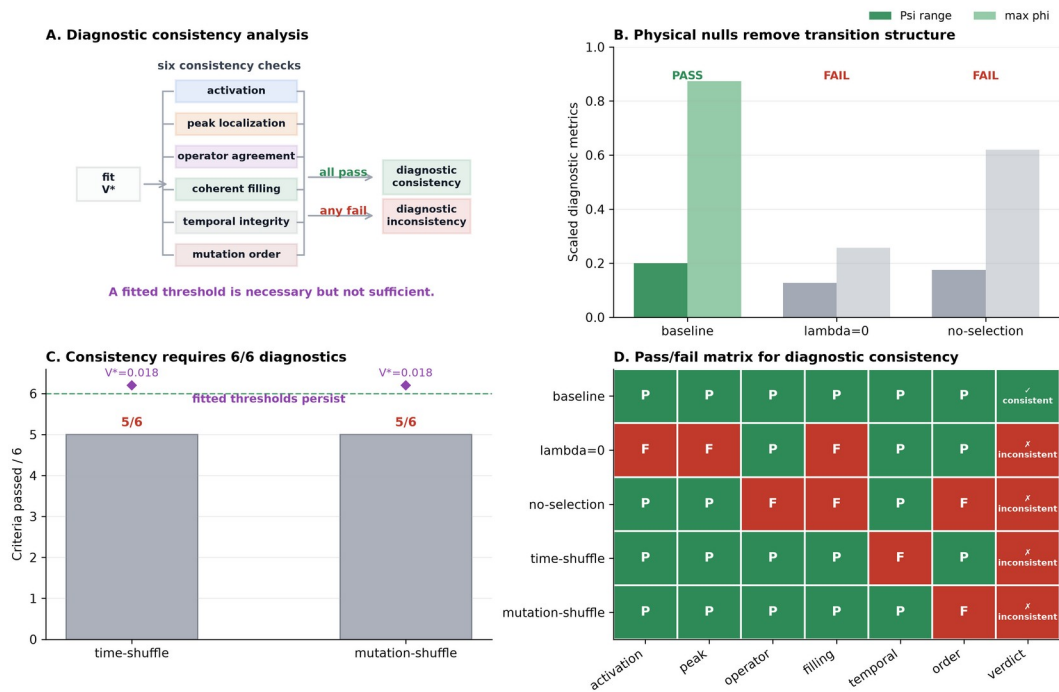


Figure 5 | Diagnostic consistency analysis

A fitted threshold alone is insufficient to establish a coherent accessibility transition. (A) Diagnostic consistency protocol. Candidate thresholds are interpreted as diagnostically consistent only when all diagnostics pass; failure of any diagnostic indicates diagnostic inconsistency. (B) Scaled diagnostic metrics for the baseline model and physical null models. Null systems can retain apparent threshold locations while losing the structural signatures required for an accessibility transition. (C) Shuffle controls illustrate threshold artifacts: fitted transition locations can persist even when temporal or mutation-order diagnostics fail. (D) Pass/fail matrix across the six diagnostics. Only the baseline system satisfies accessibility activation, peak localization, operator agreement, coherent filling, temporal integrity, and mutation-order consistency simultaneously. Together, these diagnostics provide artifact-rejection controls for distinguishing coherent accessibility activation from threshold-like fitting artifacts.

Declarations

Competing interests: The author declares no competing interests.

Data and code availability: Data and code are available in a public repository. Repository details have been withheld from the blinded manuscript and will be provided according to journal policy.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT and Codex to assist with manuscript organization, language editing, figure-production workflow management, and submission-formatting tasks. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

- [1] S. Wright, The roles of mutation, inbreeding, crossbreeding, and selection in evolution, *Proceedings of the Sixth International Congress of Genetics* 1, 356 (1932).
- [2] S. Gavrilets, *Fitness Landscapes and the Origin of Species* (Princeton University Press, Princeton, 2004).
- [3] M. Scheffer, S. Carpenter, J. A. Foley, C. Folke, and B. Walker, Catastrophic shifts in ecosystems, *Nature* 413, 591 (2001). doi:10.1038/35098000.
- [4] M. Scheffer, J. Bascompte, W. A. Brock, V. Brovkin, S. R. Carpenter, V. Dakos, H. Held, E. H. van Nes, M. Rietkerk, and G. Sugihara, Early-warning signals for critical transitions, *Nature* 461, 53 (2009). doi:10.1038/nature08227.
- [5] S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed. (Westview Press, Boulder, 2015).
- [6] J. Guckenheimer and P. Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields* (Springer, New York, 1983).
- [7] S. Wiggins, *Introduction to Applied Nonlinear Dynamical Systems and Chaos*, 2nd ed. (Springer, New York, 2003).
- [8] J. P. Sethna, K. Dahmen, S. Kartha, J. A. Krumhansl, B. W. Roberts, and J. D. Shore, Hysteresis and hierarchies: Dynamics of disorder-driven first-order phase transformations, *Physical Review Letters* 70, 3347 (1993). doi:10.1103/PhysRevLett.70.3347.

- [9] M. A. Krasnosel'skii and A. V. Pokrovskii, *Systems with Hysteresis* (Springer, Berlin, 1989).
- [10] I. D. Mayergoyz, *Mathematical Models of Hysteresis and Their Applications*, 2nd ed. (Academic Press, San Diego, 2003).
- [11] H. Mori, Transport, collective motion, and Brownian motion, *Progress of Theoretical Physics* 33, 423 (1965). doi:10.1143/PTP.33.423.
- [12] R. Zwanzig, Memory effects in irreversible thermodynamics, *Physical Review* 124, 983 (1961). doi:10.1103/PhysRev.124.983.
- [13] P. Deuffhard and M. Weber, Robust Perron cluster analysis in conformation dynamics, *Linear Algebra and its Applications* 398, 161 (2005). doi:10.1016/j.laa.2004.10.026.
- [14] C. Schutte, A. Fischer, W. Huisinga, and P. Deuffhard, A direct approach to conformational dynamics based on hybrid Monte Carlo, *Journal of Computational Physics* 151, 146 (1999). doi:10.1006/jcph.1999.6231.
- [15] M. Dellnitz and O. Junge, On the approximation of complicated dynamical behavior, *SIAM Journal on Numerical Analysis* 36, 491 (1999). doi:10.1137/S0036142996313002.
- [16] J.-H. Prinz, H. Wu, M. Sarich, B. Keller, M. Senne, M. Held, J. D. Chodera, C. Schutte, and F. Noe, Markov models of molecular kinetics: generation and validation, *Journal of Chemical Physics* 134, 174105 (2011). doi:10.1063/1.3565032.
- [17] F. Noe and F. Nueske, A variational approach to modeling slow processes in stochastic dynamical systems, *Multiscale Modeling & Simulation* 11, 635 (2013). doi:10.1137/110858616.
- [18] V. Dakos, M. Scheffer, E. H. van Nes, V. Brovkin, V. Petoukhov, and H. Held, Slowing down as an early warning signal for abrupt climate change, *Proceedings of the National Academy of Sciences* 105, 14308 (2008). doi:10.1073/pnas.0802430105.
- [19] C. Kuehn, A mathematical framework for critical transitions: bifurcations, fast-slow systems and stochastic dynamics, *Physica D* 240, 1020 (2011). doi:10.1016/j.physd.2011.02.012.
- [20] A. Hastings and D. B. Wysham, Regime shifts in ecological systems can occur with no warning, *Ecology Letters* 13, 464 (2010). doi:10.1111/j.1461-0248.2010.01439.x.

- [21] H. A. Kramers, Brownian motion in a field of force and the diffusion model of chemical reactions, *Physica* 7, 284 (1940). doi:10.1016/S0031-8914(40)90098-2.
- [22] S. A. H. Geritz, E. Kisdi, G. Meszner, and J. A. J. Metz, Evolutionarily singular strategies and the adaptive growth and branching of the evolutionary tree, *Evolutionary Ecology* 12, 35 (1998). doi:10.1023/A:1006554906681.
- [23] U. Dieckmann and R. Law, The dynamical theory of coevolution: A derivation from stochastic ecological processes, *Journal of Mathematical Biology* 34, 579 (1996). doi:10.1007/BF02409751.
- [24] J. A. J. Metz, R. M. Nisbet, and S. A. H. Geritz, How should we define “fitness” for general ecological scenarios?, *Trends in Ecology & Evolution* 7, 198 (1992). doi:10.1016/0169-5347(92)90073-K.
- [25] R. J. Bagley, J. D. Farmer, S. A. Kauffman, and N. H. Packard, Modeling adaptive biological systems, *BioSystems* 23, 113 (1989). doi:10.1016/0303-2647(89)90016-6.
- [26] P. A. Corning, A systems theory of biological evolution, *BioSystems* 214, 104630 (2022). doi:10.1016/j.biosystems.2022.104630.
- [27] J. E. Stewart, The direction of evolution: The rise of cooperative organization, *BioSystems* 123, 27 (2014). doi:10.1016/j.biosystems.2014.05.006.
- [28] S. F. Elena and R. Sanjuan, RNA viruses as complex adaptive systems, *BioSystems* 81, 31 (2005). doi:10.1016/j.biosystems.2005.02.001.
- [29] D. M. Rhodes, M. Holcombe, and E. E. Qvarnstrom, Reducing complexity in an agent based reaction model: Benefits and limitations of simplifications in relation to run time and system level output, *BioSystems* 147, 35 (2016). doi:10.1016/j.biosystems.2016.06.002.
- [30] T. Haruna, Theory of interface: Category theory, directed networks and evolution of biological networks, *BioSystems* 116, 62 (2013). doi:10.1016/j.biosystems.2013.08.002.